

[5 points]

Use mathematical induction to show that when n is an exact power of 2, the solution of the recurrence

$$T(n) = \begin{cases} 2, & \text{if } n = 2 \\ 2T(n/2) + n, & \text{if } n = 2^k, k > 1 \end{cases}$$

is

$$T(n) = n \log n$$

[5 points]

We can express insertion sort as a recursive procedure as follows:

To sort $A[1..n]$, we recursively sort $A[1..n - 1]$ and then insert $A[n]$ into the sorted array $A[1..n - 1]$.

Write a recurrence for the worst-case running time of this recursive version of insertion sort.

[10 points]

Design an algorithm to rearrange elements of a given array of n real numbers so that all its negative elements precede all its positive elements.

Your algorithm should be time- and space-efficient.

[10 points]

Find the average number of key comparisons made by binary search in a successful search in an array of size n .